

Combinational Logic Design

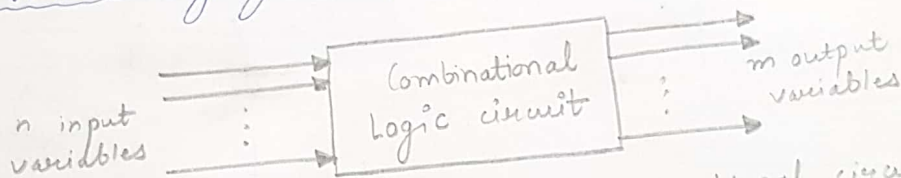
Introduction -

Logic circuits for digital systems may be -

← Combinational

→ Sequential

1. The o/p of a combinational circuit depends on its present inputs only.
2. Combinational circuit performs a specific information processing operation fully specified logically by a set of Boolean functions.
3. A combinational circuit consists of i/p variables, logic gates & o/p variables.



The Block diagram of a combinational circuit

For n i/p variables, there are 2^n possible combinations of binary i/p values.

For each possible i/p combination, there is one and only one possible o/p combination.

A combinational circuit can be described by m Boolean functions one for each o/p variable. Each o/p function is expressed in terms of n i/p variables.

Arithmetic -

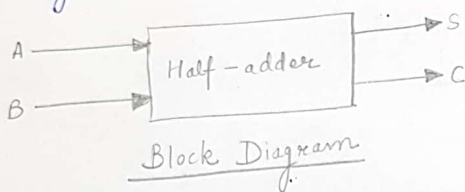
$$\begin{array}{r} A \text{ (Augend)} \\ + B \text{ (Addend)} \\ \hline \end{array}$$

$$0+0=0, 0+1=1, 1+0=1, 1+1=10$$

If A & B both are equal to 1, binary sum consists of 2 digits. The higher significant bit of this result is called a carry. When augend & addend contains next higher-order pair of significant bits, carry obtained from addition of 2 bits is added to next higher pair.

The Half-Adder -

A half-adder is a combinational circuit with two binary inputs (augend & addend bits) & two binary outputs (sum and carry bits). It adds 2 i/p's (A & B) & produces sum (S) and carry (C).

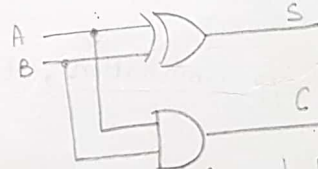


Inputs		Outputs	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Truth Table

$$S = \bar{A}B + A\bar{B} = A \oplus B$$

$$C = AB$$



Logic diagram of a half-adder

Teach

Realisation of a half-adder using Universal gates -

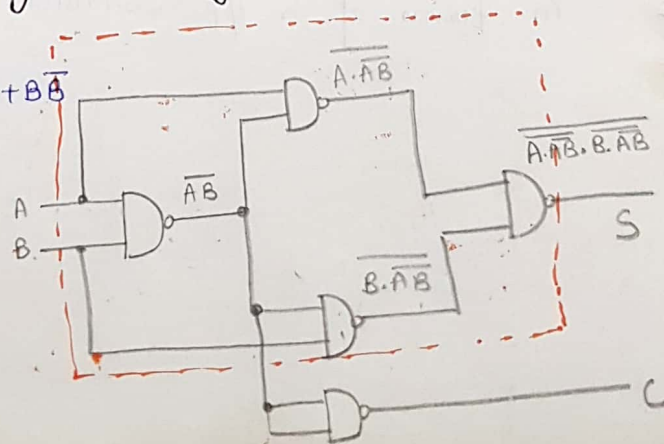
NAND Logic -

$$S = \overline{AB + \bar{A}\bar{B}} = \overline{AB} + \overline{\bar{A}\bar{B}} = A(\bar{A} + \bar{B}) + B(\bar{A} + \bar{B})$$

$$= A.\bar{A}\bar{B} + B.\bar{A}\bar{B}$$

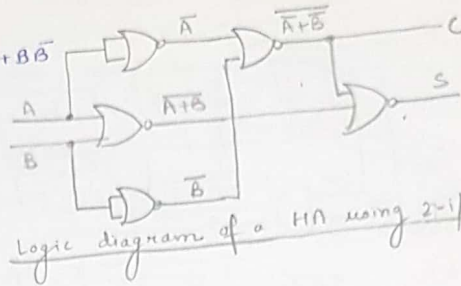
$$= \overline{A.\bar{A}\bar{B} . B.\bar{A}\bar{B}}$$

$$C = AB = \overline{\overline{AB}}$$



NOR logic -

$$\begin{aligned} S &= A\bar{B} + \bar{A}B = A\bar{B} + \bar{A}\bar{A} + \bar{A}B + B\bar{B} \\ &= A(\bar{A} + B) + \bar{A}(\bar{A} + \bar{B}) \\ &= (A + B)(\bar{A} + \bar{B}) \\ &= \overline{A+B} + \overline{\bar{A} + \bar{B}} \\ C &= AB = \overline{\bar{A}\bar{B}} = \overline{\bar{A} + \bar{B}} \end{aligned}$$



Logic diagram of a HA using 2-input NOR gate

The Full-Adder -

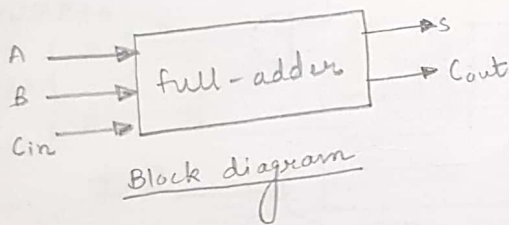
A full adder is a combinational circuit that adds 2 bits to a carry & outputs a sum bit to a carry bit.

When we want to add 2 binary nos. each having 2 or more bits, LSB can be added by using a half-adder. Carry resulted from addition of LSBs is carried over to next significant column & added to 2 bits in that column. $\therefore$, 2 bits in that column & carry bit generated from add'n in previous column need to be added.

Carry from previous column - carry-in C_{in}

Carry bit in result - carry-out C_{out}

FA adds 2 bits A & B to C_{in} to outputs sum bit S to C_{out} .



Block diagram

Inputs			Sum	Carry
A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Truth Table

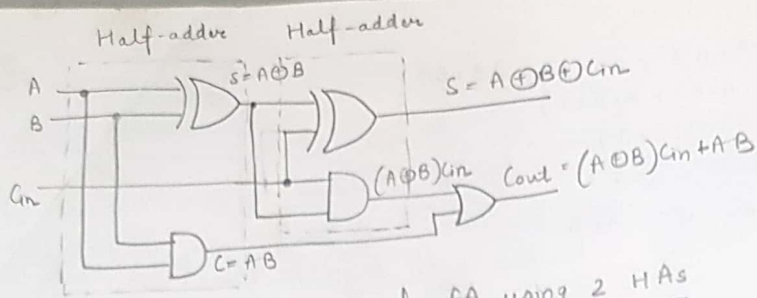
$$\begin{aligned} S &= \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in} \\ &= (\bar{A}\bar{B} + \bar{A}B)C_{in} + (A\bar{B} + \bar{A}B)C_{in} \\ &= (A \oplus B)C_{in} + (\bar{A} \oplus \bar{B})C_{in} \end{aligned}$$

$$S = A \oplus B \oplus C_{in}$$

$$\begin{aligned} C_{out} &= \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in} \\ &= \bar{A}B\bar{C}_{in} + \bar{A}B\bar{C}_{in} + \bar{A}B\bar{C}_{in} + \bar{A}B\bar{C}_{in} + \bar{A}B\bar{C}_{in} + \bar{A}B\bar{C}_{in} \\ &= \bar{A}B(C_{in} + \bar{C}_{in}) + B\bar{C}_{in}(A + \bar{A}) + A\bar{C}_{in}(B + \bar{B}) \end{aligned}$$

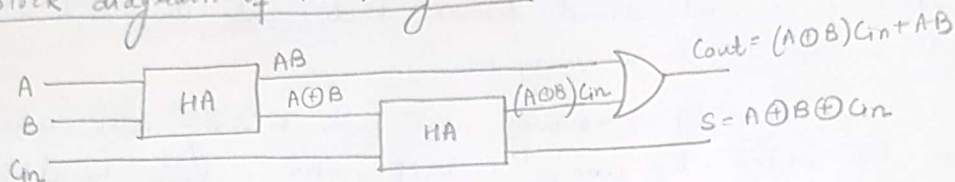
$$C_{out} = AB + B\bar{C}_{in} + A\bar{C}_{in}$$

$$\begin{aligned} \text{Also, } C_{out} &= \bar{A}B\bar{C}_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in} = \bar{A}B(C_{in} + \bar{C}_{in}) + (\bar{A}B + \bar{A}B)C_{in} \\ C_{out} &= AB + (A \oplus B)C_{in} \end{aligned}$$



Logic diagram of a FA using 2 HAs

Block diagram of FA using 2 HAs -



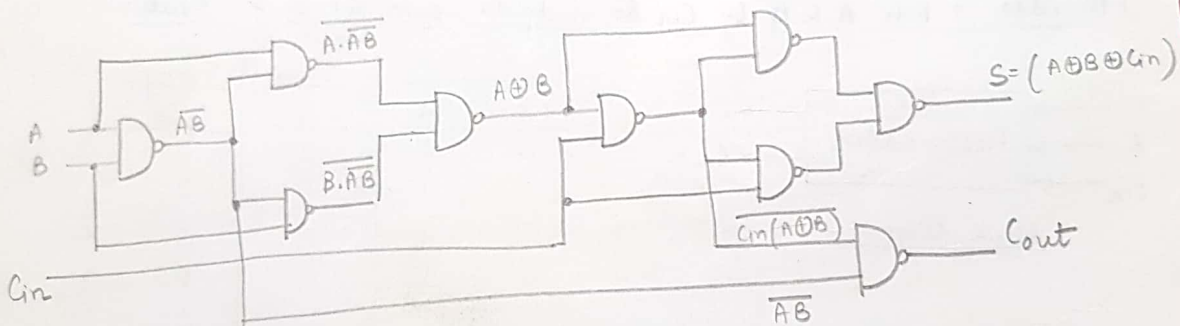
Realisation of FA using Universal gates -

NAND Logic -

Since, $A \oplus B = \overline{A \cdot AB \cdot B \cdot AB}$

$$S = A \oplus B \oplus C_{in} = \overline{(A \oplus B) \cdot (A \oplus B)C_{in} \cdot C_{in} \cdot (A \oplus B)C_{in}}$$

$$C_{out} = C_{in}(A \oplus B) + AB = \overline{C_{in}(A \oplus B) \cdot AB}$$



NOR Logic -

Since, $A \oplus B = \overline{(A+B) + \overline{A+B}}$

$$S = A \oplus B \oplus C_{in} = \overline{(A \oplus B) + C_{in} + (A \oplus B) + C_{in}}$$

$$C_{out} = AB + C_{in}(A \oplus B) = \overline{\overline{A+B} + \overline{C_{in} + A \oplus B}}$$

Implement the circuit.